

FINANCIAL CRIME FORENSIC INVESTIGATION REPORT

Case Identifier: CUSTOM-20260918-001 | Source: CUSTOM UPLOADED TRANSACTION DATASET | Generated: 2026-09-18 13:30:00 UTC

LEGAL & EVIDENTIARY DISCLAIMER: This dossier is an automated analytical investigation-support artifact produced from observed transaction logs. Inferred account roles, pattern detections, and priority rankings represent evidence-grounded hypotheses generated by mathematical graph and behavioural engines. This document does not establish criminal liability and is intended solely to guide qualified human investigators.

1. EXECUTIVE SUMMARY & FORENSIC METRICS

Monitored Accounts:	14	Total Volume (INR):	■3,285,000.00
Total Transactions:	15	Pattern Findings:	12 clusters
Attack Money Trails:	3	Top Priority Entity:	ACC_MULE3 (AccountRole.MULE - 89.0/100)
Primary DNA Signature:	DNA-LAY7-LAY7-RAP-FAN -RET82-VEL11-DISP1-H7	Typology Code:	CUSTOM_INGESTION

2. INPUT DATA & VALIDATION AUDIT

Dataset Source / Name:	JARVIS Live Custom Verification	Validation Status:	✓ VALIDATED (100% INGESTED)
Total Input Rows:	15	Valid Transactions:	15 rows
Invalid / Filtered:	0 rows	Active Entity Count:	14 accounts

3. NETWORK TOPOLOGY & COMMUNITY INTELLIGENCE

- **Network Topology:** Directed MultiGraph with 14 entity nodes and 15 transaction edges.
- **Community Partitioning:** 2 isolated / connected sub-clusters detected.
- **Dominant Flow Hubs (Inflow+Outflow):** ACC_SOURCE (■0), ACC_MULE1 (■0), ACC_MULE2 (■0), ACC_MULE3 (■0), ACC_DISPERSER (■0)

4. AML SUSPICIOUS PATTERN FINDINGS (12 Total Detections)

Pattern Type	Severity	Accounts Involved	Observable Evidence / Description
PatternType.LAYERING	PATTERNSE VERITY.HIGH	ACC_SOURCE, ACC_MULE1, ACC_MULE2, ACC_MULE3, ACC_DISPERSER	Funds moved through a sequential chain of 5 accounts: ACC_SOURCE → ACC_MULE1 → ACC_MULE2 → ACC_MULE3 → ACC_DISPERSER with high amount retention and rapid transit. [Metrics: hop_count: 4 initial_amount_inr: 500000.0 final_amount_inr: 470000.0 retention_percentage: 94.0 elapsed_transit_minutes: 25.0]
PatternType.LAYERING	PATTERNSE VERITY.MEDI UM	ACC_MULE1, ACC_MULE2, ACC_MULE3, ACC_DISPERSER	Funds moved through a sequential chain of 4 accounts: ACC_MULE1 → ACC_MULE2 → ACC_MULE3 → ACC_DISPERSER with high amount retention and rapid transit. [Metrics: hop_count: 3 initial_amount_inr: 490000.0 final_amount_inr: 470000.0 retention_percentage: 95.92 elapsed_transit_minutes: 17.0]
PatternType.CIRCULAR_ TRANSFER	PATTERNSE VERITY.CRITI CAL	ACC_COMPANY, ACC_VENDOR1	Funds circulate in a closed loop through 2 accounts: ACC_COMPANY → ACC_VENDOR1 → ACC_COMPANY, indicating potential round-tripping or fictitious settlement. [Metrics: cycle_length: 2 cycle_duration_hours: 1.0 return_percentage: 20.0 volume_inr: 30000.0]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.HIGH	ACC_SOURCE, ACC_MULE1, ACC_MULE2	High-velocity fund forwarding detected at account ACC_MULE1: ■500,000.00 received and ■490,000.00 forwarded in 8.0 minutes. [Metrics: mule_account: ACC_MULE1 inflow_amount_inr: 500000.0 outflow_amount_inr: 490000.0 dwell_time_minutes: 8.0 forwarding_ratio: 0.98 velocity_rating: ULTRA_HIGH]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.HIGH	ACC_MULE1, ACC_MULE2, ACC_MULE3	High-velocity fund forwarding detected at account ACC_MULE2: ■490,000.00 received and ■480,000.00 forwarded in 7.0 minutes. [Metrics: mule_account: ACC_MULE2 inflow_amount_inr: 490000.0 outflow_amount_inr: 480000.0 dwell_time_minutes: 7.0 forwarding_ratio: 0.98 velocity_rating: ULTRA_HIGH]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.HIGH	ACC_MULE2, ACC_MULE3, ACC_DISPERSER	High-velocity fund forwarding detected at account ACC_MULE3: ■480,000.00 received and ■470,000.00 forwarded in 10.0 minutes. [Metrics: mule_account: ACC_MULE3 inflow_amount_inr: 480000.0 outflow_amount_inr: 470000.0 dwell_time_minutes: 10.0 forwarding_ratio: 0.979 velocity_rating: HIGH]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.MEDI UM	ACC_DISPERSER, ACC_BENEFICIARY1, ACC_AGGREGATOR	High-velocity fund forwarding detected at account ACC_BENEFICIARY1: ■150,000.00 received and ■145,000.00 forwarded in 15.0 minutes. [Metrics: mule_account: ACC_BENEFICIARY1 inflow_amount_inr: 150000.0 outflow_amount_inr: 145000.0 dwell_time_minutes: 15.0 forwarding_ratio: 0.967 velocity_rating: HIGH]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.MEDI UM	ACC_DISPERSER, ACC_BENEFICIARY2, ACC_AGGREGATOR	High-velocity fund forwarding detected at account ACC_BENEFICIARY2: ■145,000.00 received and ■140,000.00 forwarded in 16.0 minutes. [Metrics: mule_account: ACC_BENEFICIARY2 inflow_amount_inr: 145000.0 outflow_amount_inr: 140000.0 dwell_time_minutes: 16.0 forwarding_ratio: 0.966 velocity_rating: HIGH]
PatternType.RAPID_MO VEMENT	PATTERNSE VERITY.MEDI UM	ACC_DISPERSER, ACC_BENEFICIARY3, ACC_AGGREGATOR	High-velocity fund forwarding detected at account ACC_BENEFICIARY3: ■140,000.00 received and ■135,000.00 forwarded in 17.0 minutes. [Metrics: mule_account: ACC_BENEFICIARY3 inflow_amount_inr: 140000.0 outflow_amount_inr: 135000.0 dwell_time_minutes: 17.0 forwarding_ratio: 0.964 velocity_rating: HIGH]
PatternType.FAN_IN	PATTERNSE VERITY.MEDI UM	ACC_BENEFICIARY2, ACC_BENEFICIARY3, ACC_BENEFICIARY1, ACC_AGGREGATOR	Concentration of funds detected: 3 accounts routed a combined total of ■420,000.00 into aggregator ACC_AGGREGATOR. [Metrics: aggregator_account: ACC_AGGREGATOR feeder_count: 3 transaction_count: 3 total_aggregated_inr: 420000.0 time_span_hours: 0.07]
PatternType.FAN_OUT	PATTERNSE VERITY.MEDI UM	ACC_DISPERSER, ACC_BENEFICIARY2, ACC_BENEFICIARY3, ACC_BENEFICIARY1	Dispersion pattern detected: Single account ACC_DISPERSER split and transferred ■435,000.00 across 3 distinct recipient accounts. [Metrics: disperser_account: ACC_DISPERSER beneficiary_count: 3 transaction_count: 3 total_dispersed_inr: 435000.0 time_span_hours: 0.03]

Pattern Type	Severity	Accounts Involved	Observable Evidence / Description
PatternType.FAN_OUT	PATTERNSEVERITY.MEDIUM	ACC_COMPANY, ACC_VENDOR1, ACC_VENDOR2, ACC_VENDOR3	Dispersion pattern detected: Single account ACC_COMPANY split and transferred █75,000.00 across 3 distinct recipient accounts. [Metrics: disperser_account: ACC_COMPANY beneficiary_count: 3 transaction_count: 3 total_dispersed_inr: 75000.0 time_span_hours: 2.0]

5. RECONSTRUCTED ATTACK PATHS & MONEY TRAILS (3 Active Trails)

Path ID	Flow Route (Source → Intermediaries → Sink)	Hops	Initial Vol	Retention	Duration
PATH_001	ACC_SOURCE → ACC_MULE1 → ACC_MULE2 → ACC_MULE3 → ACC_DISPERSER → ACC_BENEFICIARY1 → ACC_AGGREGATOR → ACC_SINK	7	█0	82.0%	0 min
PATH_002	ACC_SOURCE → ACC_MULE1 → ACC_MULE2 → ACC_MULE3 → ACC_DISPERSER → ACC_BENEFICIARY2 → ACC_AGGREGATOR → ACC_SINK	7	█0	82.0%	0 min
PATH_003	ACC_SOURCE → ACC_MULE1 → ACC_MULE2 → ACC_MULE3 → ACC_DISPERSER → ACC_BENEFICIARY3 → ACC_AGGREGATOR → ACC_SINK	7	█0	82.0%	0 min

6. PROBABLE ACCOUNT ROLE HYPOTHESES

Account ID	Probable Role	Conf.	Total Inflow	Total Outflow	Forwarding	Dwell Time
ACC_SOURCE	AccountRole.ORIGINATOR	92%	█0	█500,000	0%	0 min
ACC_MULE1	AccountRole.MULE	86%	█500,000	█490,000	98%	0 min
ACC_MULE2	AccountRole.MULE	86%	█490,000	█480,000	98%	0 min
ACC_MULE3	AccountRole.MULE	86%	█480,000	█470,000	98%	0 min
ACC_DISPERSER	AccountRole.DISPERSER	88%	█470,000	█435,000	93%	0 min
ACC_BENEFICIARY1	AccountRole.MULE	86%	█150,000	█145,000	97%	0 min
ACC_BENEFICIARY2	AccountRole.MULE	86%	█145,000	█140,000	97%	0 min
ACC_BENEFICIARY3	AccountRole.MULE	86%	█140,000	█135,000	96%	0 min
ACC_AGGREGATOR	AccountRole.AGGREGATOR	89%	█420,000	█410,000	98%	0 min
ACC_SINK	AccountRole.SINK	90%	█410,000	█0	0%	0 min
ACC_COMPANY	AccountRole.DISPERSER	88%	█5,000	█75,000	100%	0 min
ACC_VENDOR1	AccountRole.MULE	65%	█25,000	█5,000	20%	0 min
ACC_VENDOR2	AccountRole.SINK	90%	█18,000	█0	0%	0 min
ACC_VENDOR3	AccountRole.SINK	90%	█32,000	█0	0%	0 min

7. INVESTIGATION PRIORITY TRIAGE (0–100 EXPLAINABLE SCORING)

Rank	Account ID	Priority Score	Inferred Role	Confidence	Observable Evidence Rationale
#-	ACC_MULE3	89.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_DISPERSER	89.0/100	AccountRole.DISPERSER	0%	Flow volume and centrality
#-	ACC_MULE2	87.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_MULE1	83.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_AGGREGATOR	80.0/100	AccountRole.AGGREGATOR	0%	Flow volume and centrality
#-	ACC_SOURCE	60.0/100	AccountRole.ORIGINATOR	0%	Flow volume and centrality
#-	ACC_BENEFICIARY1	58.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_BENEFICIARY2	57.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_BENEFICIARY3	57.0/100	AccountRole.MULE	0%	Flow volume and centrality
#-	ACC_COMPANY	37.0/100	AccountRole.DISPERSER	0%	Flow volume and centrality

8. MONEY TRAIL DNA™ BEHAVIOURAL FINGERPRINT

Active DNA Fingerprint: [DNA-LAY7-LAY7-RAP-FAN-RET82-VEL11-DISP1-H7](#)
SHA-256 Evidence Seal: cce982a73fb3df148b9b7fbd6ff59bb6f3f0582b30e6e86e76b3d06c52a1bf63

Gene Identifier	Gene Code	Value / Signal	Forensic Calculation / Meaning
Typology	LAY7-LAY7-RAP-FAN	LAY7 → LAY7 → RAP → FAN	Reconstructed money flow manifests 4 distinct AML pattern typologies across 7 transit hops.
Retention	RET82	82.0% Retained	Initial path disbursement was █500,000.00 and final terminal receipt was █410,000.00, resulting in a 82.0% fund retention rate (█90,000.00 transit fee/leakage).
Velocity	VEL11	10.8 mins (HIGH)	Average dwell latency between intermediary transfer hops is 10.8 minutes (HIGH velocity forwarding).
Dispersion	DISP1	1 → 6 → 1	Fan-out branching factor of 1 outgoing channels converging across 6 intermediate nodes into 1 feeder streams.
Topology	H7	7 Hops (8 Accounts)	Graph path depth spanning 7 sequential transfers across 8 distinct financial entities.
Role Sequence	ORI-MUL-MUL-MUL-DIS-MUL-AGG-SIN	ORIGINATOR → MULE → MULE → MULE → DISPERSER → MULE → AGGREGATOR → SINK	Inferred operational path topology progressing through: ORIGINATOR → MULE → MULE → MULE → DISPERSER → MULE → AGGREGATOR → SINK.

9. HISTORICAL CASE BEHAVIOURAL SIMILARITY

Automated vector distance matching against historical typologies from law enforcement repositories:

Historical Reference Case	Similarity	Primary Typology	Matching Behavioural Vectors / Signals
None — None	0.0%	N/A	Behavioural match: Typology and velocity alignment
None — None	0.0%	N/A	Behavioural match: Typology and velocity alignment
None — None	0.0%	N/A	Behavioural match: Typology and velocity alignment
None — None	0.0%	N/A	Behavioural match: Typology and velocity alignment

10. TEMPORAL RISK EVOLUTION & STORYLINE

Stage	Stage Name	Time Interval	Tx Count	Volume (INR)	Observed Typology Phase
#-	INJECTION	2026-09-18T09:00 to 2026-09-18T09:00	-	■0	
#-	DISPERSION	2026-09-18T09:08 to 2026-09-18T11:00	-	■0	
#-	LAYERING	2026-09-18T09:25 to 2026-09-18T13:00	-	■0	
#-	CONVERGENCE	2026-09-18T09:45 to 2026-09-18T09:49	-	■0	
#-	SINK	2026-09-18T10:05 to 2026-09-18T10:05	-	■0	

Chronological Event Sequence:

- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:
- Event:

11. COMPREHENSIVE INVESTIGATOR NARRATIVE

11.1 Executive Synthesis:

JARVIS-AML investigation completed for CUSTOM-20260918-001. The network encompasses 14 monitored accounts and 15 transactions totaling ■3,285,000.00. Analysis detected 12 suspicious pattern clusters (RAPID_MOVEMENT, FAN_IN, CIRCULAR_TRANSFER, LAYERING, FAN_OUT). The highest-priority money trail spans 7 hops from ACC_SOURCE to ACC_SINK, transferring ■500,000.00 with 82.0% retention over 65.0 minutes. The generated Money Trail DNA fingerprint is 'DNA-LAY7-LAY7-RAP-FAN-RET82-VEL11-DISP1-H7'. Behavioural similarity analysis identifies an 58.8% alignment with historical reference 'Micro-Smurfing Structuring Syndicate' (CASE-021).

13. TRANSACTION EVIDENCE REGISTER (KEY OBSERVATIONS)

Showing chronological transaction logs extracted for forensic chain of custody (15 total records):

Tx ID	Timestamp	Sender Account	Receiver Account	Amount (INR)	Channel
TX001	2026-09-18T09:00:00	ACC_SOURCE	ACC_MULE1	■500,000	NEFT
TX002	2026-09-18T09:08:00	ACC_MULE1	ACC_MULE2	■490,000	IMPS
TX003	2026-09-18T09:15:00	ACC_MULE2	ACC_MULE3	■480,000	IMPS
TX004	2026-09-18T09:25:00	ACC_MULE3	ACC_DISPERSER	■470,000	NEFT
TX005	2026-09-18T09:30:00	ACC_DISPERSER	ACC_BENEFICIARY1	■150,000	UPI

Tx ID	Timestamp	Sender Account	Receiver Account	Amount (INR)	Channel
TX006	2026-09-18T09:31:00	ACC_DISPERSER	ACC_BENEFICIARY2	■145,000	UPI
TX007	2026-09-18T09:32:00	ACC_DISPERSER	ACC_BENEFICIARY3	■140,000	UPI
TX008	2026-09-18T09:45:00	ACC_BENEFICIARY1	ACC_AGGREGATOR	■145,000	NEFT
TX009	2026-09-18T09:47:00	ACC_BENEFICIARY2	ACC_AGGREGATOR	■140,000	NEFT
TX010	2026-09-18T09:49:00	ACC_BENEFICIARY3	ACC_AGGREGATOR	■135,000	NEFT
TX011	2026-09-18T10:05:00	ACC_AGGREGATOR	ACC_SINK	■410,000	RTGS
TX012	2026-09-18T11:00:00	ACC_COMPANY	ACC_VENDOR1	■25,000	NEFT
TX013	2026-09-18T11:30:00	ACC_COMPANY	ACC_VENDOR2	■18,000	NEFT
TX015	2026-09-18T13:00:00	ACC_COMPANY	ACC_VENDOR3	■32,000	UPI
TX014	2026-09-18T12:00:00	ACC_VENDOR1	ACC_COMPANY	■5,000	NEFT

14. FORENSIC CHAIN OF CUSTODY & INTEGRITY SEAL

CRYPTOGRAPHIC EVIDENCE INTEGRITY SEAL	Generated: 2026-09-18 13:30:00 UTC
Case Evidence SHA-256 Digest:	cce982a73fb3df148b9b7fbd6ff59bb6f3f0582b30e6e86e76b3d06c52a1bf63
Verification Protocol:	JARVIS-AML Deterministic Graph Hash (HMAC-SHA256)